In [1]:

```python
import pandas as pd
import matplotlib.pyplot as plt
from pathlib import Path
import os
from pathlib import Path
```

Descriptive Statistics Section

GDP

In [2]:

```python
df_Final_data_path = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df_Final_data = pd.read_excel(df_Final_data_path / "final_panel_dataset.xlsx")
df_Final = df_Final_data[['country', 'year', 'gdppc']]

periods = {
    '1882-1913': (1882, 1913),
    '1929-1971' : (1929, 1971),
    '1974-2020' : (1974, 2020)
}

results = []
for country in df_Final['country'].unique():
    df_c = df_Final[df_Final['country'] == country]

    row = {
        "Country": country,
        "Avg": df_c['gdppc'].mean(),
        "Min": df_c['gdppc'].min(),
        "Max": df_c['gdppc'].max()
    }

    # Add averages for each sub-period
    for label, (start, end) in periods.items():
        if label == "Avg":
            continue  # already computed
        mask = (df_c['year'] >= start) & (df_c['year'] <= end)
        row[label] = df_c.loc[mask, 'gdppc'].mean()

    results.append(row)

df_gdp_table = pd.DataFrame(results)

df_gdp_table = df_gdp_table.sort_values("Country").reset_index(drop=True)


df_gdp_table
```

Out[2]:

	Country	Avg	Min	Max	1882-1913	1929-1971	1974-2020
0	Australia	18181.714409	5451.0000	49814.669956	6878.343750	12220.511628	34459.942613
1	Canada	17379.594154	3304.0000	45284.213482	4652.812500	11967.860465	34199.352923
2	Denmark	17659.407083	3571.0000	48238.657869	4705.593750	11910.023256	35084.820948
3	Finland	13614.012432	1918.0000	40129.653300	2498.062500	7908.325581	29559.653788
4	France	15180.038225	3518.0000	39084.655420	4315.093750	9560.348837	30561.964111
5	Germany	15595.977800	3258.0000	46761.946056	4433.062500	9514.488372	31987.147111
6	Italy	13382.222730	2831.0000	36310.734900	3302.437500	7458.348837	28501.573606
7	Japan	13110.189602	1723.5686	38414.861800	2042.516029	5877.909449	30262.578890
8	Netherlands	17120.066007	4222.0000	48142.671760	5447.781250	10772.813953	33915.152658
9	Norway	21341.497371	2499.0000	85115.323086	3150.656250	9376.465116	50162.470949
10	Portugal	8766.608128	1581.0000	28056.056110	1862.593750	4145.441860	19730.585741
11	Spain	10539.900708	2330.0000	34957.465528	2661.812500	4826.906977	23397.110603
12	Sweden	16214.236925	2354.0000	45434.402948	3216.000000	11387.511628	32823.828353
13	Switzerland	20955.820699	3998.6616	62448.599363	6185.244103	13535.309081	41794.540964
14	United Kingdom	16398.416105	6132.0000	39113.011733	7148.531250	12140.465116	29177.358267
15	United States	22284.062755	6424.1432	56469.263659	7927.131712	15936.648963	41479.469879

Descripitve Stats Labour Productivity

In [3]:

```python
df_Final_data_path = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df_Final_data = pd.read_excel(df_Final_data_path / "final_panel_dataset.xlsx")
df_Final = df_Final_data[['country', 'year', 'labour_productivity']]

periods = {
    '1882-1913': (1882, 1913),
    '1929-1971' : (1929, 1971),
    '1974-2020' : (1974, 2020)
}
results = []
for country in df_Final['country'].unique():
    df_c = df_Final[df_Final['country'] == country]

    row = {
        "Country": country,
        "Avg": df_c['labour_productivity'].mean(),
        "Min": df_c['labour_productivity'].min(),
        "Max": df_c['labour_productivity'].max()
    }

    # Add averages for each sub-period
    for label, (start, end) in periods.items():
        if label == "Avg":
            continue  # already computed
        mask = (df_c['year'] >= start) & (df_c['year'] <= end)
        row[label] = df_c.loc[mask, 'labour_productivity'].mean()

    results.append(row)

df_lp_table = pd.DataFrame(results)
df_lp_table = df_lp_table.sort_values("Country").reset_index(drop=True)
df_lp_table
```

Out[3]:

	Country	Avg	Min	Max	1882-1913	1929-1971	1974-2020
0	Australia	24.412274	7.145286	57.693508	8.844059	17.531114	42.896392
1	Canada	23.736365	3.617792	59.533274	4.965576	16.858045	44.322236
2	Denmark	25.806922	3.550943	71.875021	4.742646	14.321432	52.951044
3	Finland	19.752029	1.930385	60.669460	2.515039	8.764999	43.751207
4	France	23.766845	2.756078	67.915392	3.276825	10.839148	52.094069
5	Germany	25.141762	4.585027	68.980046	6.014343	12.461173	52.173867
6	Italy	21.375320	2.018100	55.582627	2.378297	9.819482	47.023313
7	Japan	15.411422	0.970538	49.062035	1.324780	5.463195	35.819434
8	Netherlands	28.735044	5.239417	68.968044	5.925238	17.139801	56.715737
9	Norway	34.941293	3.641319	97.520531	4.286882	17.039040	75.829314
10	Portugal	14.053882	1.892896	38.907656	2.293712	6.964283	29.851783
11	Spain	20.032883	3.888716	53.113263	4.565802	8.862955	42.525098
12	Sweden	24.500651	2.619799	68.115160	3.722773	14.473205	49.860418
13	Switzerland	34.285447	5.536116	80.029640	7.020768	25.147592	63.801251
14	United Kingdom	23.198715	5.962112	59.286516	6.602303	13.397534	45.323436
15	United States	28.725957	4.626473	73.957562	5.621785	21.188733	53.373439

Descriptive Stats CPI

In [4]:

```python
df_Final_data_path = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df_Final_data = pd.read_excel(df_Final_data_path / "final_panel_dataset.xlsx")
df_Final = df_Final_data[['country', 'year', 'cpi']]

periods = {
    '1882-1913': (1882, 1913),
    '1929-1971' : (1929, 1971),
    '1974-2020' : (1974, 2020)
}

results = []
for country in df_Final['country'].unique():
    df_c = df_Final[df_Final['country'] == country]

    row = {
        "Country": country,
        "Avg": df_c['cpi'].mean(),
        "Min": df_c['cpi'].min(),
        "Max": df_c['cpi'].max()
    }

    # Add averages for each sub-period
    for label, (start, end) in periods.items():
        if label == "Avg":
            continue  # already computed
        mask = (df_c['year'] >= start) & (df_c['year'] <= end)
        row[label] = df_c.loc[mask, 'cpi'].mean()

    results.append(row)

df_cpi_table = pd.DataFrame(results)
df_cpi_table = df_cpi_table.sort_values("Country").reset_index(drop=True)
df_cpi_table
```

Out[4]:

	Country	Avg	Min	Max	1882-1913	1929-1971	1974-2020
0	Australia	43.544890	2.250000e+00	202.046014	2.682292e+00	9.146283	116.351147
1	Canada	45.877413	5.285346e+00	174.814103	6.240752e+00	15.904300	112.104647
2	Denmark	41.964554	2.109956e+00	161.815857	2.395390e+00	10.352823	110.231615
3	Finland	38.950207	4.620000e-02	162.232694	5.784375e-02	6.985444	107.648813
4	France	39.327044	6.050080e-02	155.324987	6.281631e-02	8.350713	107.488288
5	Germany	51.007171	1.150362e-11	168.789036	1.271163e-11	30.324740	118.420254
6	Italy	43.403157	2.208800e-02	200.084964	2.349424e-02	4.744669	123.435780
7	Japan	37.840419	1.604590e-02	111.607085	2.478576e-02	12.560825	98.654745
8	Netherlands	50.393610	5.380884e+00	181.424326	6.084630e+00	19.351761	121.941864
9	Norway	43.073465	2.293544e+00	187.737714	2.555589e+00	10.715319	112.724314
10	Portugal	45.323773	4.173523e-02	226.825215	5.211314e-02	2.170814	131.626799
11	Spain	44.773411	2.568748e-01	213.584964	3.040568e-01	3.726903	128.092626
12	Sweden	41.697069	2.407773e+00	161.660079	2.784503e+00	10.395410	109.000505
13	Switzerland	50.689569	9.257564e+00	132.437189	1.062807e+01	28.040517	108.051906
14	United Kingdom	43.987101	1.984892e+00	194.847983	2.110475e+00	8.816898	118.340113
15	United States	49.896845	6.736232e+00	192.622537	7.368184e+00	19.124733	119.524709

Descriptive Stats - Trade Openness

In [5]:

```python
df_Final_data_path = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df_Final_data = pd.read_excel(df_Final_data_path / "final_panel_dataset.xlsx")
df_Final = df_Final_data[['country', 'year', 'trade_openness']]

periods = {
    '1882-1913': (1882, 1913),
    '1929-1971' : (1929, 1971),
    '1974-2020' : (1974, 2020)
}
results = []
for country in df_Final['country'].unique():
    df_c = df_Final[df_Final['country'] == country]

    row = {
        "Country": country,
        "Avg": df_c['trade_openness'].mean(),
        "Min": df_c['trade_openness'].min(),
        "Max": df_c['trade_openness'].max()
    }

    # Add averages for each sub-period
    for label, (start, end) in periods.items():
        if label == "Avg":
            continue  # already computed
        mask = (df_c['year'] >= start) & (df_c['year'] <= end)
        row[label] = df_c.loc[mask, 'trade_openness'].mean()

    results.append(row)

df_to_table = pd.DataFrame(results)
df_to_table = df_to_table.sort_values("Country").reset_index(drop=True)
df_to_table
```

Out[5]:

	Country	Avg	Min	Max	1882-1913	1929-1971	1974-2020
0	Australia	2.417701e-01	1.422638e-01	4.240048e-01	2.292362e-01	2.196393e-01	2.877856e-01
1	Canada	4.049497e-04	2.345392e-04	6.922896e-04	3.493633e-04	3.198211e-04	5.070648e-04
2	Denmark	4.844653e-04	1.122112e-04	6.341054e-04	4.699919e-04	4.164150e-04	5.435144e-04
3	Finland	2.016736e+00	4.574830e-01	2.996546e+00	2.465905e+00	2.001051e+00	1.599314e+00
4	France	1.970264e-03	4.171811e-04	3.428153e-03	1.877088e-03	1.254822e-03	2.546570e-03
5	Germany	2.545903e-02	1.109059e-04	9.217843e-01	6.141566e-04	3.630010e-04	9.946653e-04
6	Italy	5.772409e-01	9.202175e-02	1.163041e+00	4.112590e-01	3.749590e-01	8.930931e-01
7	Japan	2.078527e-07	1.282452e-08	3.809018e-07	1.806363e-07	1.924190e-07	2.129923e-07
8	Netherlands	1.670221e+00	2.418701e-01	2.744003e+00	2.183510e+00	1.131527e+00	2.089003e+00
9	Norway	4.247826e-01	1.296565e-01	5.558980e-01	3.936148e-01	3.624605e-01	4.984302e-01
10	Portugal	3.620249e-01	8.010407e-02	8.677099e-01	9.682671e-02	2.982708e-01	6.349794e-01
11	Spain	2.502477e-01	3.128278e-02	6.700941e-01	1.517355e-01	1.211439e-01	4.716263e-01
12	Sweden	3.827747e-01	1.143868e-01	6.744272e-01	3.491635e-01	2.816653e-01	5.266224e-01
13	Switzerland	5.286803e-01	1.634748e-01	1.416275e+00	6.925174e-01	3.504960e-01	5.804088e-01
14	United Kingdom	3.659191e-04	1.011181e-04	5.634698e-04	4.673845e-04	2.725937e-04	3.636764e-04
15	United States	1.393867e-04	4.984706e-05	3.084248e-04	1.060479e-04	7.968947e-05	2.247321e-04

Descriptive Stats - liquid Liabiliteis

In [6]:

```python
df_Final_data_path = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df_Final_data = pd.read_excel(df_Final_data_path / "final_panel_dataset.xlsx")
df_Final = df_Final_data[['country', 'year', 'liquid_liabilities']].copy()

periods = {
    'Avg 1882-1913': (1882, 1913),
    'Avg 1929-1971': (1929, 1971),
    'Avg 1974-2020': (1974, 2020)
}

results = []
for country in sorted(df_Final['country'].unique()):
    df_c = df_Final[df_Final['country'] == country]

    row = {
        "Country": country,
        "Avg": df_c['liquid_liabilities'].mean(),
        "Min": df_c['liquid_liabilities'].min(),
        "Max": df_c['liquid_liabilities'].max()
    }
    for label, (start, end) in periods.items():
        mask = (df_c['year'] >= start) & (df_c['year'] <= end)
        row[label] = df_c.loc[mask, 'liquid_liabilities'].mean()

    results.append(row)


df_liq_table = pd.DataFrame(results)
df_liq_table = df_liq_table.sort_values("Country").reset_index(drop=True)

df_liq_table
```

Out[6]:

	Country	Avg	Min	Max	Avg 1882-1913	Avg 1929-1971	Avg 1974-2020
0	Australia	0.588035	0.378306	1.219731	0.475271	0.564270	0.723271
1	Canada	0.438497	0.210120	0.971409	0.335605	0.412538	0.537484
2	Denmark	0.496879	0.324765	0.749691	0.577666	0.452793	0.451458
3	Finland	0.499915	0.242607	0.823183	0.497352	0.480582	0.510755
4	France	0.661069	0.287009	2.259868	0.651115	0.636227	0.626257
5	Germany	0.512251	0.057906	1.131326	0.601928	0.289691	0.644742
6	Italy	0.630216	0.304372	1.235554	0.444598	0.642336	0.761780
7	Japan	0.735814	0.209509	2.030276	0.287033	0.641858	1.206171
8	Netherlands	0.551982	0.199094	1.685238	0.255081	0.584656	0.762501
9	Norway	0.596664	0.388443	1.107734	0.578366	0.618113	0.524919
10	Portugal	0.511492	0.087546	1.168655	0.121681	0.510912	0.855755
11	Spain	0.608370	0.207639	1.262809	0.277856	0.603928	0.908790
12	Sweden	0.624740	0.385460	0.875163	0.593502	0.701271	0.548066
13	Switzerland	0.926073	0.481506	1.616489	0.567522	1.013753	1.142984
14	United Kingdom	0.589564	0.410412	1.183510	0.470901	0.571418	0.716867
15	United States	0.529296	0.291136	0.846065	0.399205	0.598438	0.559694

Descriptive Stats --> Stock Market Cap

In [7]:

```python
df_Final_data_path = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df_Final_data = pd.read_excel(df_Final_data_path / "final_panel_dataset.xlsx")
df_Final = df_Final_data[['country', 'year', 'mcap_gdp']]
periods = {
    '1882-1913': (1882, 1913),
    '1929-1971' : (1929, 1971),
    '1974-2020' : (1974, 2020)
}
results = []
for country in df_Final['country'].unique():
    df_c = df_Final[df_Final['country'] == country]

    row = {
        "Country": country,
        "Avg": df_c['mcap_gdp'].mean(),
        "Min": df_c['mcap_gdp'].min(),
        "Max": df_c['mcap_gdp'].max()
    }

    # Add averages for each sub-period
    for label, (start, end) in periods.items():
        if label == "Avg":
            continue  # already computed
        mask = (df_c['year'] >= start) & (df_c['year'] <= end)
        row[label] = df_c.loc[mask, 'mcap_gdp'].mean()

    results.append(row)
df_smc_table = pd.DataFrame(results)
df_smc_table = df_smc_table.sort_values("Country").reset_index(drop=True)
df_smc_table
```

Out[7]:

	Country	Avg	Min	Max	1882-1913	1929-1971	1974-2020
0	Australia	0.418503	0.132369	1.395296	0.208414	0.290691	0.703213
1	Canada	0.714749	0.268621	1.409225	0.469164	0.819846	0.800997
2	Denmark	0.289178	0.070842	1.220458	0.337489	0.142894	0.426838
3	Finland	NaN	NaN	NaN	NaN	NaN	NaN
4	France	0.396188	0.054870	0.967841	0.611027	0.274747	0.429348
5	Germany	0.239331	0.024887	0.714247	0.287374	0.147618	0.308788
6	Italy	NaN	NaN	NaN	NaN	NaN	NaN
7	Japan	NaN	NaN	NaN	NaN	NaN	NaN
8	Netherlands	NaN	NaN	NaN	NaN	NaN	NaN
9	Norway	0.209818	0.029932	0.832498	0.208851	0.109701	0.307934
10	Portugal	0.176810	0.004320	0.793364	0.078213	0.163738	0.253180
11	Spain	0.226914	0.055092	0.729985	0.080589	0.199170	0.348403
12	Sweden	0.382406	0.074692	1.409290	0.257910	0.211559	0.648783
13	Switzerland	0.679780	0.157315	2.797097	0.313177	0.400360	1.310204
14	United Kingdom	0.774818	0.198062	1.881063	0.706594	0.720806	0.951829
15	United States	0.674530	0.252113	1.534410	0.575888	0.561486	0.889553

Descriptive Stats --> Banks Loans

In [8]:

```python
df_Final_data_path = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df_Final_data = pd.read_excel(df_Final_data_path / "final_panel_dataset.xlsx")
df_Final = df_Final_data[['country', 'year', 'bank_loans_gdp']]
periods = {
    '1882-1913': (1882, 1913),
    '1929-1971' : (1929, 1971),
    '1974-2020' : (1974, 2020)
}

results = []
for country in df_Final['country'].unique():
    df_c = df_Final[df_Final['country'] == country]

    row = {
        "Country": country,
        "Avg": df_c['bank_loans_gdp'].mean(),
        "Min": df_c['bank_loans_gdp'].min(),
        "Max": df_c['bank_loans_gdp'].max()
    }

    # Add averages for each sub-period
    for label, (start, end) in periods.items():
        if label == "Avg":
            continue  # already computed
        mask = (df_c['year'] >= start) & (df_c['year'] <= end)
        row[label] = df_c.loc[mask, 'bank_loans_gdp'].mean()

    results.append(row)

df_bl_table = pd.DataFrame(results)
df_bl_table = df_bl_table.sort_values("Country").reset_index(drop=True)
df_bl_table
```

Out[8]:

	Country	Avg	Min	Max	1882-1913	1929-1971	1974-2020
0	Australia	0.456235	0.160158	1.606247	0.395751	0.225695	0.785653
1	Canada	0.469079	0.141676	1.124800	0.429858	0.263298	0.719027
2	Denmark	1.130445	0.547949	2.035726	1.028275	0.951024	1.347576
3	Finland	0.443136	0.086383	0.989226	0.295511	0.344873	0.642041
4	France	NaN	NaN	NaN	NaN	NaN	NaN
5	Germany	0.630132	0.050678	1.040243	0.743309	0.374665	0.844116
6	Italy	0.438776	0.128907	0.999568	0.207631	0.397922	0.679816
7	Japan	0.576814	0.107728	1.542900	0.257677	0.488570	0.869021
8	Netherlands	NaN	NaN	NaN	NaN	NaN	NaN
9	Norway	0.720617	0.229805	1.498531	0.651111	0.537671	0.891203
10	Portugal	NaN	NaN	NaN	NaN	NaN	NaN
11	Spain	NaN	NaN	NaN	NaN	NaN	NaN
12	Sweden	0.681006	0.376188	1.301502	0.644173	0.478102	0.910948
13	Switzerland	1.123329	0.647735	1.676905	0.952047	1.058768	1.308577
14	United Kingdom	0.421918	0.136823	1.182136	0.259581	0.250274	0.751273
15	United States	0.421697	0.115199	0.695372	0.329269	0.339959	0.555112

In [9]:

```python
df_Final_data_path = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df_Final_data = pd.read_excel(df_Final_data_path / "final_panel_dataset.xlsx")
df_Final = df_Final_data[['country', 'year', 'gdppc', 'labour_productivity', 'cpi', 'trade_openness', 'liquid_liabilities', 'mcap_gdp', 'bank_loans_gdp']]

df_Final.describe()
```

Out[9]:

	year	gdppc	labour_productivity	cpi	trade_openness	liquid_liabilities	mcap_gdp	bank_loans_gdp
count	2224.000000	2224.000000	2096.000000	2.224000e+03	2.224000e+03	2224.000000	1488.000000	1668.000000
mean	1951.000000	16107.735321	24.242301	4.448436e+01	4.052065e-01	0.593804	0.430062	0.626099
std	40.133829	14479.989948	21.727817	5.785014e+01	6.466756e-01	0.257660	0.372890	0.359545
min	1882.000000	1581.000000	0.970538	1.150362e-11	1.282452e-08	0.057906	0.004320	0.050678
25%	1916.000000	4997.250000	6.242636	2.666016e+00	4.547730e-04	0.431571	0.167840	0.341700
50%	1951.000000	9841.075950	14.581455	1.192234e+01	1.717040e-01	0.548365	0.321797	0.553202
75%	1986.000000	24807.509900	40.460965	8.554003e+01	4.825879e-01	0.707641	0.578042	0.838633
max	2020.000000	85115.323086	97.520531	2.268252e+02	2.996546e+00	2.259868	2.797097	2.035726

Looking for Discontinutiy

In [10]:

```python
import pandas as pd

# Load your data (adjust path if needed)
df_Final_data_path = Path("/Users/lorenzouberti/Desktop/Panel-Data-Project-Financial-Econometrics/Data")
df = pd.read_excel(df_Final_data_path / "final_panel_dataset.xlsx")

# Make sure data are sorted by country and year
df = df.sort_values(["country", "year"])

def count_panel_stats(group):
    years = group["year"].tolist()
    n_obs = len(years)

    # Count gaps > 1 year
    discontinuities = sum(
        (years[i] - years[i-1] > 1)
        for i in range(1, n_obs)
    )

    # Count consecutive pairs (year_t - year_(t-1) == 1)
    consecutive_pairs = sum(
        (years[i] - years[i-1] == 1)
        for i in range(1, n_obs)
    )

    return pd.Series({
        "n_obs": n_obs,
        "discontinuities": discontinuities,
        "consecutive_pairs": consecutive_pairs
    })

# Apply function by country
panel_stats = df.groupby("country").apply(count_panel_stats)

# Display results
print(panel_stats)
```

```
                n_obs  discontinuities  consecutive_pairs
country
Australia         139                0                138
Canada            139                0                138
Denmark           139                0                138
Finland           139                0                138
France            139                0                138
Germany           139                0                138
Italy             139                0                138
Japan             139                0                138
Netherlands       139                0                138
Norway            139                0                138
Portugal          139                0                138
Spain             139                0                138
Sweden            139                0                138
Switzerland       139                0                138
United Kingdom    139                0                138
United States     139                0                138
/var/folders/k1/mhyzg0w15gb04d0x6q7gdp2w0000gn/T/ipykernel_62663/3757882027.py:33: DeprecationWarning: DataFrameGroupBy.apply operated on the grouping columns. This behavior is deprecated, and in a f
uture version of pandas the grouping columns will be excluded from the operation. Either pass `include_groups=False` to exclude the groupings or explicitly select the grouping columns after groupby t
o silence this warning.
  panel_stats = df.groupby("country").apply(count_panel_stats)
```

Time Series Plot of the Explanatory Variable Part 1 Question 2